

Full length article

A novel real-time aluminum-copper dissimilar metal ARM laser welding seam width at the interface monitoring method based on multi-diagnostic signals

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ABSTRACT

Aluminum alloys and copper alloys are typical low-resistance metals, which are widely used in the electrical industry. The adjustable-ring-mode (ARM) laser welding, with its advantages of small heat-affected zone, fast welding speed, and almost no spatter, is an ideal method for aluminum-copper dissimilar metal joining. Due to the differences in physical properties such as melting point and thermal conductivity between aluminum and copper, the seam width at the interface (SWI) tends to fluctuate significantly, which affects the electrical and mechanical properties of the joint. Therefore, monitoring SWI is an important method for evaluating the aluminum-copper dissimilar metal joint quality. In this study, a novel real-time aluminum-copper dissimilar metal SWI monitoring method based on optical coherence tomography (OCT) keyhole depth signals and plasma plume spectral signal in ARM laser welding is proposed. Based on the analysis of correlations of several features on the cross-section, a method for characterizing the SWI based on the upper and lower materials melted volumes and the penetration depth was proposed. Signal processing techniques are applied to denoise and analyze OCT and spectral signals, confirming the strong correlation between multiple signals and key features in the SWI characterizing model, which can be used for SWI prediction. Finally, based on multi-signal diagnosis, a back-propagation neural network (BPNN) model for SWI prediction is established. The results show that the average error of this method is only 10.7 μm , achieving high-precision prediction of SWI in aluminum-copper dissimilar metal ARM laser welding.

1. Introduction

Copper is a typical metal with low electrical resistance and high thermal conductivity [1]. Aluminum alloy is valued for its relatively high electrical conductivity and lightweight [2]. They are commonly used materials in industries such as shipbuilding, aerospace, and automotive manufacturing for electrical components [3]. For instance, in the production of power batteries for new energy vehicles, aluminum and copper are often used as raw materials for battery tabs and busbars, connected in joint form [4–7]. Various methods are commonly used for aluminum-copper joining, including friction stir welding, ultrasonic welding, magnetic pulse welding, resistance spot welding, electron beam welding, and laser welding [8]. Among them, ARM laser welding is considered an ideal method for dissimilar metal joining due to its

advantages of high welding speed, high energy density, high depth-to-width ratio, small heat-affected zone, and minimal spatter [9–11]. However, due to the significant differences in physical properties such as melting point, thermal conductivity, and laser absorption rate between aluminum alloys and copper alloys, and the dynamic of the molten pool and keyhole during laser deep penetration welding, the geometric dimensions of the melt pool, including the SWI, fluctuate dramatically [12–14]. Therefore, monitoring the welding process is crucial to ensure the quality of the final product [15]. As a result, many researchers have studied methods for quality inspection using visual cameras, acoustic signals, ultrasonic testing, optical information, and other techniques [16].

The most researched method is using visual cameras to capture photos or videos of the melt pool or metal vapor during the welding

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process for welding quality monitoring [17,18]. As early as 1997, J. Beersiek et al. used a CCD camera to capture images of the keyhole and melt pool, achieving $\pm 10\%$ accuracy in penetration depth monitoring based on the relationship between keyhole and melt pool size and penetration depth [19]. In 2000, Connolly, J.O. et al. established a laser welding monitoring system based on visible light wavelength optical signals, enabling the classification and prediction of stainless-steel laser welding penetration depth [20]. In 2012, Fanrong Kong et al. used a high-speed CCD camera assisted by green laser to monitor the dynamic behavior of the molten pool, finding a strong correlation between metal vapor and keyhole dynamics, thereby achieving real-time monitoring of joint formation dimensions during laser welding process [21]. In 2019, Wei Meng et al. utilized images of the molten pool and plasma plume obtained through two high-speed camera to calculate instantaneous features such as plasma plume area, peak frequency, and melt pool length, and discovered a strong correlation between these features and joint gap [22]. In 2022, Wang Cai et al. proposed a strategy for diagnosing penetration status based on two deep learning neural networks and image processing, ultimately achieving a highly robust real-time prediction of penetration status in laser deep penetration welding with an accuracy of up to 98.68% [23,24]. However, the limited capability of visual cameras to capture only narrow wavelength range external light signals restricts their application range in monitoring welding process features inside the joints during laser welding.

To obtain broader spectrum and higher precision optical radiation information during laser welding processes, many researchers have attempted to monitor the laser welding process by collecting spectra of plasma plumes above the melt pool [25,26]. T. Sibillano et al. initiated a series of studies on monitoring and controlling the penetration depth of stainless-steel laser welding based on spectral signals starting from 2010. Initially, they calculated the corresponding electron temperature of the plasma plume and found a decrease in electron temperature and an increase in penetration depth with increasing laser power [27]. They proposed that this decrease in electron temperature with increasing penetration depth might caused by the lower position of the high-temperature core region of the plasma in the keyhole during deep penetration, resulting in a lower detected plasma temperature. Subsequently, based on the quantitative relationship between plasma electron temperature and penetration depth, they achieved penetration depth monitoring of stainless steel laser welding [28]. Finally, they established a penetration depth monitoring and control platform for stainless steel laser welding, enabling closed-loop control of penetration depth [29]. Jiyoung Yu et al. collected spectral signals of plasma plumes in the range of 225–975 nm during laser welding processes, extracted spectral signal features by transforming all collected spectral signals into the CIE1931RGB color space, and achieved a prediction accuracy of 90% for penetration status [30]. Shixuan Li et al. found that certain characteristic spectral lines in plasma spectra could reflect the penetration state between dissimilar metal joints and the upper and lower plate fusion states, and based on their relative intensities, they achieved penetration depth prediction with an error of ± 0.05 mm [31]. The intensity information of various characteristic spectral lines and electron temperature information of the plasma are highly correlated with the penetration state of the base material or dynamic behavior of the keyhole during welding processes. However, plasma spectra, as an external signal of laser welding keyholes, have limited capability to characterize internal weld features.

In recent years, many researchers have attempted to use OCT technology for monitoring the dynamic of keyholes depth in order to obtain more direct information inside the molten pool [32]. From 2007 to 2010, Paul J. L. Webster et al. pioneered the use of OCT technology in the field of laser welding and ultrafast laser ablation process monitoring. They integrated OCT devices into welding heads, enabling real-time coaxial monitoring of the welding process, which they named In-process Coherent Imaging (ICI). Based on this technology, they founded the Laser Depth Dynamics (LDD) company, laying the foundation

for the subsequent application of OCT devices in the field of laser welding [33–36]. Deyuan Ma et al. used OCT devices to capture fluctuations in keyhole depth during laser welding processes. Based on the observation that large fluctuations in keyhole depth often led to porosity defects in aluminum welds, they achieved real-time monitoring of porosity defects in aluminum alloy laser welding processes [37]. Devices utilizing OCT technology can effectively capture fluctuations in keyhole depth during laser welding processes, enabling the collection of internal features of the molten pool. However, this method can only provide information related to penetration depth, porosity defects, and other factors related to keyhole depth, and cannot monitor the SWI in laser welding joints.

Although SWI is an important indicator affecting the quality of welds in welding, there has been very little research on monitoring the SWI laser lap welding. Schimon, Ron W, and others conducted research on the relationship between the SWI and the width of the weld pool. They proposed a method for predicting the SWI based on the width of the weld surface and three-dimensional modeling of the weld [38]. N. Kumar et al. constructed a three-dimensional finite element model of laser welding welds and pointed out that this method can be used to predict the depth and width of welds. They also proposed that the size of the weld is not only determined by the line energy during the welding process but also related to welding time, cooling time, and energy density [39]. Their research has made significant contributions to the monitoring SWI. However, because their research involves modeling predictions or non-quantitative relationship analysis, it cannot achieve quantitative monitoring of SWI during laser welding processes.

This paper addresses the aforementioned issues by analyzing the relevant characteristic information of SWI and establishing a geometric model for SWI. It utilizes OCT to obtain depth-related information and spectroscopic signals to capture information related to the ratio of molten volumes between the upper and lower plates. Finally, based on BPNN and multiple signals (OCT and spectroscopic signals), the paper establishes a monitoring model for the SWI of aluminum-copper dissimilar metal ARM laser welding joints, achieving high-precision monitoring of SWI at the joint interface.

2. Experimental setup

The materials and equipment used in the experiments of this study are schematically represented in Fig. 1. The laser used is a continuous fiber laser, model RFL-4000/2000-ABP, manufactured by Wuhan Raycus Fiber Laser Technologies Co. Ltd. It can output a maximum power of 6 kW with a wavelength of 1080 ± 5 nm. The laser can produce both point and ring mode laser beams, with individually adjustable point and ring laser powers. The point spot is circular with a radius of 0.05 mm and the energy distribution is Gaussian, with a maximum power of 4 kW, adjustable every 40 W. The ring beam size is circular with a radius of 0.15 mm, with a maximum power of 2 kW, adjustable every 20 W. The laser head is fixed on EMP-10180 three-axis linear motion platform, maintaining a vertical position. It can move within a spatial range of 1800 mm*1000 mm*350 mm at a speed not exceeding 120 m/min. Spectral signals are collected using an Avantes spectrometer, model AvaSpec-UL32048Cl-6-EVO, equipped with six channels covering a wavelength range from 200 nm to 1100 nm, with channel resolutions ranging from 0.1 nm to 0.28 nm. The keyhole depth signal is acquired through an Optical Coherence Tomography (OCT) device from Lessmüller, which offers three scanning functions—pre-weld seam morphology, in-weld keyhole depth, and post-weld seam formation—all simultaneously usable. 17,000 scans can be freely set in one second, with customizable trajectories (in a range of 98.5 mm*67 mm*267 mm) and functions.

The materials used in the experiment are 1060-H24 aluminum alloy and TU1 copper alloy, with compositions and contents detailed in Table 1. Both materials are of specifications 100 mm*150 mm*1mm. The experiment involves laser overlapping welding of aluminum and

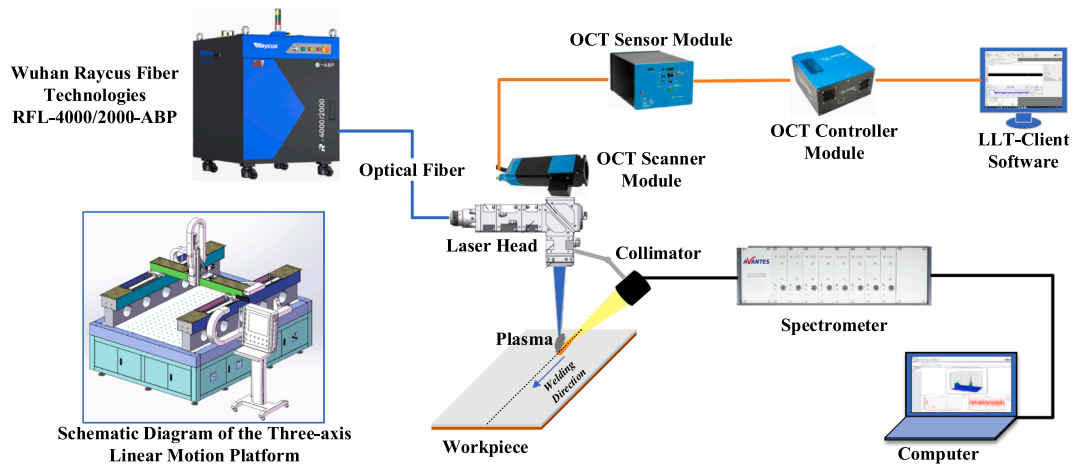


Fig. 1. Adjustable-ring-mode laser welding and monitoring platform.

Table 1
Experimental parameters.

Laser Power (W)	Welding Speed (mm/s)	Defocus (mm)	Shielding Gas Flow Rate (L/min)
1440, 1520, 1600, 1680, 1760, 1840, 1920, 2000, 2080	70	−1	5

copper, with aluminum positioned above copper. The experimental weld seam is a straight line with a total length of 100 mm, maintaining a 5° angle between the plate and the horizontal plane to ensure a 5° push angle for laser welding. To achieve optimal surface formation, the following parameters remain constant throughout the experiment: the laser point and ring power ratio is 1:1, welding speed is 70 mm/s, defocus amount is −1 mm (the melting point of TU1 is 1083 °C, which is much higher than the 660 °C of 1060, a defocus of −1 mm prevents the situation where the upper aluminum plate is fully penetrated while the lower plate has not formed a keyhole), and pure argon gas is used as the shielding gas with a flow rate of 5 L/min (To minimize the impact of the shielding gas on the quality of the spectral signals while ensuring proper weld formation). See (Table 2).

For a diverse range of SWI samples, the total laser power ranges from the lowest 1440 W (just melting the lower plate) in increments of 80 W, up to the highest 2080 W (fully melting through the lower plate). Each power level experiment is repeated 5 times to reduce accidental errors.

The collection head of the spectrometer and the laser head are fixed through a bracket, ensuring the collection head is angled downward at 45° from the welding direction to capture the plasma spectrum throughout the entire welding process. To obtain more spectra during the welding process, the frame sampling time of the spectrometer is set at 8 ms. Including the processing time needed for each spectrometer sample, the actual sampling time is around 8.35 ms. This allows for approximately 165 frames of valid welding process spectra throughout the entire welding process.

To obtain a more reliable signal for keyhole depth during the welding process, both pre-weld and in-weld Optical Coherence Tomography (OCT) scanning functions are employed. The pre-weld scanning

trajectory is a length of 4 mm, perpendicular to the welding direction and symmetrically positioned about the straight line where the weld seam is located. In-weld scanning is performed in a point scanning mode directly at the position of the keyhole. The brightness threshold in the OCT software is set to 180 to filter out points that characterize the keyhole depth.

Post-weld, there is minimal spattering on the weld seam surface, and the surface formation is satisfactory, meeting the requirements for weld seam surface formation in the context of welding battery terminals to busbars in power battery scenarios.

Through the described process, a total of 45 welds were obtained, corresponding to 9 different sets of welding parameters. To analyze the variation in their SWI and related features, 7 cross-sectional samples of each weld were taken (315 cross-sections in total), each sample measuring 2 mm*4 mm*4 mm. For the extraction of high-precision features such as weld cross-sectional width, penetration depth, and upper and lower plate melt areas, seven samples from the same weld were embedded on a metallographic sample. All cross-sectional samples were sequentially grinding using 400-grit, 800-grit, and 1200-grit sandpaper. The grinding pressure was set at 10 N, with each grinding session lasting 10 min. After grinding, the samples were polished with alumina suspension for 5 min per session, using a polishing pressure of 5 N. At last, the metallographic sample was immersed in Keller's reagent (95 ml water, 2.5 ml nitric acid, 1.5 ml hydrochloric acid, 1 ml hydrofluoric acid) for approximately 20 s, rendering the melted region of the weld clearly visible (Fig. 2).

3. Relevant features of SWI and multi-sensing signals

This Section aims to understand the relationship between process parameters, internal characteristics, and external signals, facilitating the establishment of a reliable model for monitoring the SWI in aluminum-copper dissimilar metal laser overlapping welding joints.

3.1. Analysis of geometrically relevant features and the establishment of a geometrical model for SWI

To extract high-precision feature dimensions of the weld melt zone

Table 2
Chemical composition value in samples.

Element (1060)	Al	Si	Fe	Cu	Mn	Mg	Zn	Ti			
Actual Value (wt%)	99.65	0.08	0.16	0.01	0.02	0.004	0.01	0.01			
Element (TU1)	Cu+Ag	Pb	Zn	Fe	P	Sb	O	Sn	Bi	Ni	S
Actual Value (wt%)	99.97	0.003	0.003	0.004	0.002	0.002	0.002	0.002	0.001	0.002	0.004

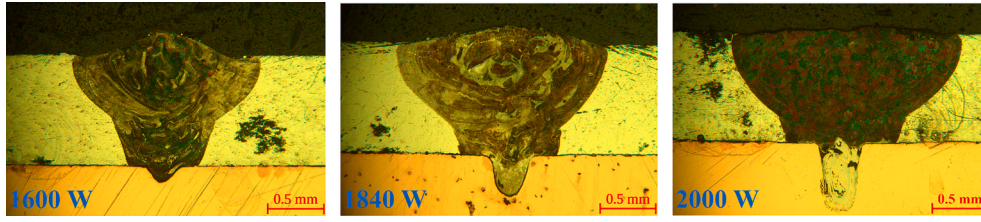


Fig. 2. The cross-sectional interface between the melted zone and the base material is clearly discernible after grinding, polishing, and corrosion under different welding power conditions.

and analyze the correlation between different features and SWI, it is necessary to denoise and analyze the scanned images of the corroded cross-sectional weld (to ensure that the molten regions of the upper and lower plates can be extracted from any cross-section, the following example uses a cross-section with a higher amount of interference (such as porosity) to demonstrate the effectiveness of the molten region extraction method). As the interface between the melted zone and the base material is clear after corrosion (Fig. 3(a)), grayscale convolution denoising was applied to the image using a 10×10 kernel (Fig. 3(b)). Subsequently, color difference analysis was performed sequentially in the horizontal and vertical directions on the optical microscope scan images of the weld cross-section (Fig. 3(c) – (d)). The images after color difference analysis enable easy distinction of dimensions for different features (In Fig. 3(e), A_1 and A_2 represent the molten areas of the upper and lower plates within the cross-section, respectively. D_1 and D_2 denote the molten depths of the upper and lower plates within the cross-section, respectively, while SWI refers to the seam width at the interface).

By analyzing the variation in different feature dimensions under various processes, it was observed that as laser power increased, the SWI exhibited an increasing trend with a decreasing rate of change. Other feature dimensions also showed an increasing trend, but with varying rates of change. Calculating the Pearson correlation coefficients between different feature dimensions and the change in SWI revealed that the volume ratio of the melted zone (copper plate melted zone/aluminum plate melted zone) and the penetration depth exhibited relatively strong correlations with the change in SWI. See (Table 3).

To establish a specific relationship between the SWI and closely related feature dimensions (volume ratio of the melted zone, penetration depth), a morphological analysis was conducted on the cross-sectional view of aluminum-copper dissimilar metal laser overlapping welding joints.

Under the same welding parameters, the melted volume remains relatively constant, while the penetration depth exhibits certain fluctu-

ations during the welding process. When the penetration depth changes, the SWI tends to increase with a decrease in the lower plate's penetration depth and increase with an increase in the lower plate's penetration depth. Under different welding parameters, significant variations were observed in the volume ratio of the melted zone, penetration depth, and SWI. A clear trend of increased SWI was observed with an increase in the volume ratio of the melted zone. Therefore, a simplified morphological model for the SWI of aluminum-copper dissimilar metal laser overlapping welding joints can be represented by the following formula (in which A_1 and A_2 represent the molten areas of the upper and lower plates within the cross-section, respectively. D_1 and D_2 denote the molten depths of the upper and lower plates within the cross-section, respectively, while SWI refers to the seam width at the interface).

$$SWI = k_1 \frac{A_2/A_1}{D_2} = k_1 \frac{A_2/A_1}{D - D_1} \quad (1)$$

To validate the reliability of the formula based on morphological patterns, a correlation analysis was conducted on the variation patterns of the ratio of A_1/A_2 to $D - D_1$ (Fig. 4 (a)) under different process parameters against SWI (Fig. 4 (b)). The resulting Pearson correlation coefficient between $\frac{A_1/A_2}{D - D_1}$ and SWI, after normalization, was as high as 0.9343. This indicates the feasibility of monitoring SWI through signals related to the volume ratio of the melted zone and penetration depth. However, due to the clear non-linear quantitative relationship between $\frac{A_1/A_2}{D - D_1}$ and SWI, achieving the goal of monitoring SWI requires further quantification and solution of their relationship.

3.2. Correlation analysis of spectral signals with plasma particle concentration and melt volume

In the welding process, metal particles existing in atomic states in the metal vapor absorb energy under the radiation of the laser. This causes

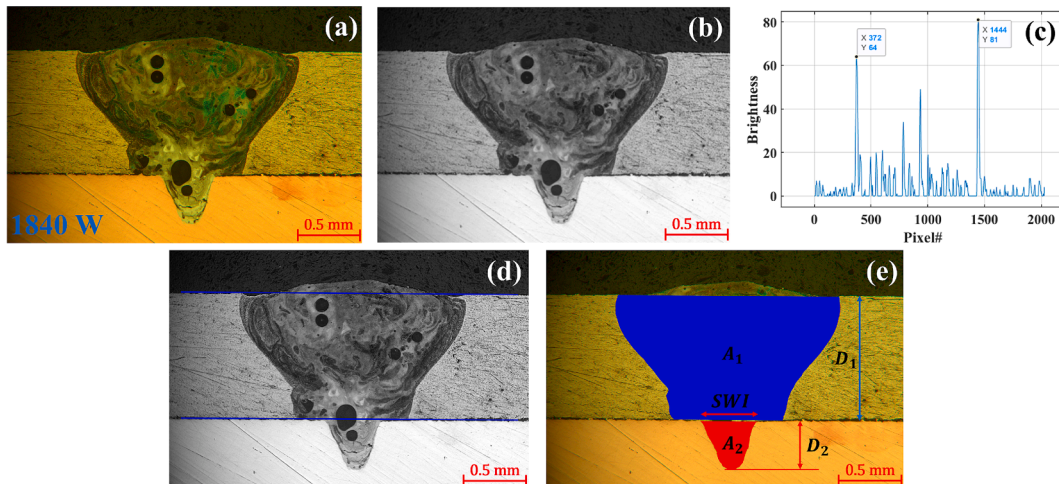
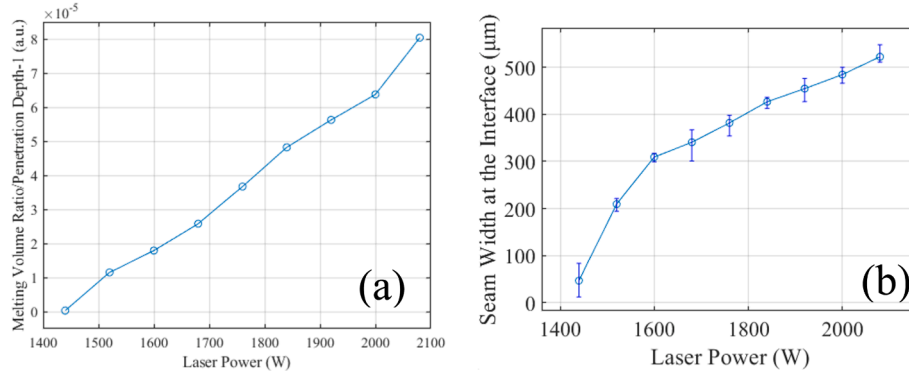
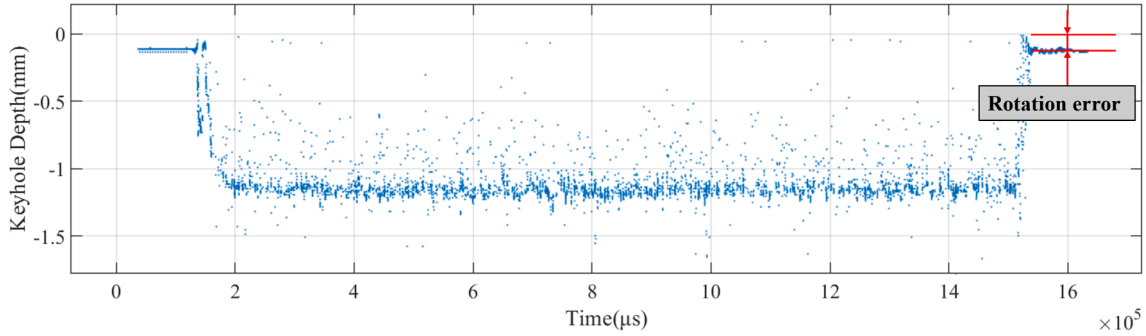


Fig. 3. (a)The scan of cross section after corrosion (b) Grayscale convolution processed scan picture (c)The vertical pixel brightness of the denoised picture (d)The edge of upper and lower plates (e)The melting zone of upper and lower plates.

Table 3

The basic parameter of aluminum and copper related continuous characteristic spectral lines [41].

	Ritz Wavelength(nm)	$A_{ji}(s^{-1})$	$E_i(cm^{-1})$	$E_j(cm^{-1})$	Lower Level Conf., Term, J	Upper Level Conf., Term, J
Al I	308.2151	5.87e + 07	0	32435.453	3s ² 3p ² P° 1/2	3s ² 3d ² D 3/2
Al I	309.27084	7.29e + 07	112.061	32436.796	3s ² 3p ² P° 3/2	3s ² 3d ² D 5/2
Al I	394.40060	4.99e + 07	0	2537.756	3s ² 3p ² P° 1/2	3s ² 4s ² S 1/2
Al I	396.15201	9.85e + 07	112.061	25347.756	3s ² 3p ² P° 3/2	3s ² 4s ² S 1/2
Cu I	324.75358	1.395e + 08	0	30783.697	3d ¹⁰ 4s ² S 1/2	3d ¹⁰ 4p ² P° 3/2
Cu I	327.39521	1.376e + 08	0	30535.324	3d ¹⁰ 4s ² S 1/2	3d ¹⁰ 4p ² P° 1/2
Cu I	578.2126	1.65e + 06	13245.443	30535.324	3d ⁹ 4s ² 2D3/2	3d ¹⁰ 4p ² P° 1/2

**Fig. 4.** (a)The changes of $\frac{A_1/A_2}{D-D_1}$ under different laser power (b) The variation of SWI with laser power.**Fig. 5.** The raw OCT signal.

electrons in the atomic nucleus to transition to higher, unstable energy levels, forming a plasma. During the process of electrons returning to more stable energy levels, they emit light of varying wavelengths based on the different energy level differences. The more frequently a specific transition occurs per unit time, the stronger the radiation intensity at that wavelength. A spectrometer's grating disperses the collected plasma radiation onto different channels of a CCD at different diffraction angles corresponding to different wavelengths. The CCD's individual capacitors convert the radiation intensity of a specific narrow bandwidth into electrical signals, which are ultimately stored in digital form in the spectrometer's RAM. Therefore, the spectra collected during the experiment can, to some extent, reflect the particle intensities of different particles in the metal vapor and plasma. According to the Boltzmann formula, assuming the photo-induced plasma is optically thin and in a state of local thermal equilibrium, its electromagnetic radiation intensity can be expressed using the following formula [40]:

$$I = \frac{n_s A_{ij} g_i \cdot \exp\left(-\frac{E_i}{kT}\right)}{Z} \quad (2)$$

in which I is spectral intensity, respectively, E_i and g_i are the energy and

the degeneracy of the upper energy level, A_{ij} is the transition probability, Z is the partition function, k is the Boltzmann constant, T is the electron temperature, n_s is the particle concentration in the laser-induced plasma [31]. In this state, the intensity of certain characteristic spectral lines (I) in the spectrum is significantly influenced by the corresponding particle concentration in the plasma. However, because electrons of an element in the plasma randomly transition to different energy levels, they emit electromagnetic radiation of various wavelengths when transitioning to more stable energy levels. To determine the concentration of an element in the plasma through spectroscopy, it is necessary to compare the obtained intensity of specific characteristic spectral lines under relatively stable conditions with the spectra of these lines under different particle concentrations. In this study, achieving an ideal instantaneous, stable state in the experimental environment proved challenging.

Under conditions of dissimilar metal lap welding, variations in the molten volumes of the top and bottom plates can lead to changes in the concentrations of the corresponding elements in the plasma plume, resulting in variations in the intensities of the characteristic spectral lines. Therefore, the ratio of the intensities of different characteristic spectral lines can indicate the ratio of the molten volumes of the top and bottom plates in dissimilar metal butt welding. To obtain information

about the upper and lower plate melt volume ratio in the laser welding process through spectral signals in this experimental environment, it is necessary to acquire multiple characteristic spectral lines corresponding to the same element to minimize the impact of environmental instability on the intensity of characteristic spectral lines.

3.3. The theory of optical coherence tomography and its relationship to penetration depth

Laser welding is primarily divided into laser conduction welding and laser deep penetration welding, with the main distinction being the presence of the keyhole effect. When the laser power is sufficiently high, the material surface rapidly heats under laser beam irradiation, reaching the boiling point in an extremely short time, leading to material melting and vaporization, creating a keyhole. The keyhole absorbs nearly all incident laser energy, reaching an equilibrium temperature inside the cavity of around 2500 °C. Heat is then transferred from the high-temperature keyhole walls, causing the surrounding metal to melt as the keyhole moves with the laser during the welding process.

Based on this phenomenon, Optical Coherence Tomography (OCT) devices are integrated coaxially in the welding head. During welding, these devices rapidly send several columns of fixed-frequency light into the keyhole. The interference results from their spatial meeting provide information on the distance the light travels from emission to reflection, representing the keyhole-to-OCT device distance. Combining this with the pre-weld measurement of the distance from the workpiece surface to the OCT device allows real-time measurement of keyhole depth during welding. This method, being a direct measurement, is minimally affected by plasma and welding fumes, offering high precision and strong resistance to interference.

However, because this method measures keyhole depth fluctuations, and there is a molten layer between the keyhole and the base material, the measured keyhole depth by OCT (D_{OCT}) often underestimates the actual penetration depth (D). Moreover, due to the significant fluctuations in keyhole depth during welding, the OCT signal may not continuously reach the bottom of the keyhole when coaxial with the laser, leading to considerable signal fluctuations and increased noise. Therefore, OCT signals need to undergo denoising processing to serve as a reliable indicator of penetration depth fluctuations.

4. Multi-signal processing and SWI monitoring model establishment

From the previous analysis, it is evident that there is a strong correlation between the molten volume ratio (A_2/A_1) and certain characteristic spectral lines in the spectrum signal (I_{sp}). Additionally, there is a significant correlation between the penetration depth (D) and the keyhole depth obtained by OCT (D_{OCT}). Considering that the molten depth of the upper plate equals one in the case of its full penetration, it can be further expressed using the detectable signals in the following form.

$$SWI = k_1 \frac{A_2/A_1}{D - D_1} = k_1 \frac{k_2 I_{sp}}{k_3 D_{OCT} - 1} = k \frac{I_{sp}}{D_{OCT} - 1} \quad (3)$$

4.1. The noise reduction and extraction of characteristic spectral lines from the spectrum signal

The spectrometer collects the sum of all radiations in the measured wavelength range (200 nm–1100 nm) for the measured region (the plasma region above the keyhole in this study). The raw results obtained from this measurement cannot be directly used for analysis and prediction. To proceed with further research, it is necessary to preprocess the spectrum signal by reducing noise. In this study, not all intensities in the 200 nm–1100 nm wavelength range are indicative of the welding state. Therefore, after noise reduction, effective analysis and selection of

characteristic spectral lines related to the molten volume ratio (A_2/A_1) are required.

4.1.1. The denoising of the multi-channel spectral signal is based on the secondary lower envelope line separation

The plasma spectrum collected during the welding process not only includes the intensity of characteristic spectral lines that represent the melting volume of the upper and lower plates but also contains background radiation, molecular spectral lines, absorption spectral lines, and other forms of radiation. These radiations are widely distributed across various wavelength bands, and their superimposed form generally results in the collected spectral intensity being higher than that of the characteristic spectral lines, as shown by the black line in Fig. 6. Due to the use of a multi-channel spectrometer in this study (indicated by green vertical lines in Fig. 6), spatial acquisition errors exist among different channels, causing significant differences in the radiation intensity corresponding to different wavelengths, further increasing the signal noise at each wavelength in the spectrum.

To separate the intensity of characteristic spectral lines applicable to this study from the collected spectrum, denoising of the spectral signal is required. In contrast to the uniformly distributed noise in the spectrum, the intensity of individual characteristic spectral lines is often highly concentrated, leading to a sharp increase and decrease in intensity within an extremely narrow band (corresponding to 1 ~ 3 pixels width on the photosensitive element of the spectrometer) and presenting a needle-like shape on the spectrum graph. Therefore, using the secondary lower envelope line within a single channel (indicated by the red line in Fig. 6) can effectively separate the spectral intensity of non-characteristic spectral lines. Subsequently, combining the denoised signals from each channel provides the denoised spectral characteristic line signals for the entire wavelength band, as illustrated by the blue line in Fig. 6, providing more accurate data for the characteristic spectral lines in subsequent studies.

4.1.2. Sliding window-based feature spectral line extraction

Based on denoising, the spectral signal now contains information about all lines in the 200 nm–1100 nm wavelength range, which includes the intensity of radiation emitted by particles such as aluminum, copper, magnesium, chromium, and aluminum oxide radicals. To select characteristic spectral lines related to aluminum and copper for characterizing the melting volume ratio (A_2/A_1), a sliding window approach was applied to filter the spectral signal. Comparing the wavelengths of the selected 67 characteristic spectral lines with the NIST database and previous studies, 65 of them were associated with radiation sources. The intensities of these characteristic spectral lines are not stable during the welding process. To identify stable characteristic spectral lines during welding, the distribution of characteristic spectral lines on the signal's time axis was analyzed for each frame. Only characteristic spectral lines present in over 70 % of all frames corresponding to the welding process were retained to ensure that the selected characteristic spectral lines could adequately represent the element's content in the plasma during welding.

In Fig. 7, each blue dot represents a characteristic spectral line present in that frame. The blue dots connected by the red line represent characteristic spectral lines with good continuity during the welding process in this study. Finally, from the 20 characteristic spectral lines related to aluminum and copper, 7 well-continuous characteristic spectral lines (Fig. 8) were selected as the basis for characterizing the melting volume ratio (A_2/A_1).

4.2. OCT signal denoising based on empirical mode decomposition

In order to prevent laser reflection from high-reflectivity metals like aluminum and copper, a 5° tilt angle was set in this experiment to ensure that the reflected laser does not hit the welding head. This also led to a deviation between the OCT welding front base surface height detection

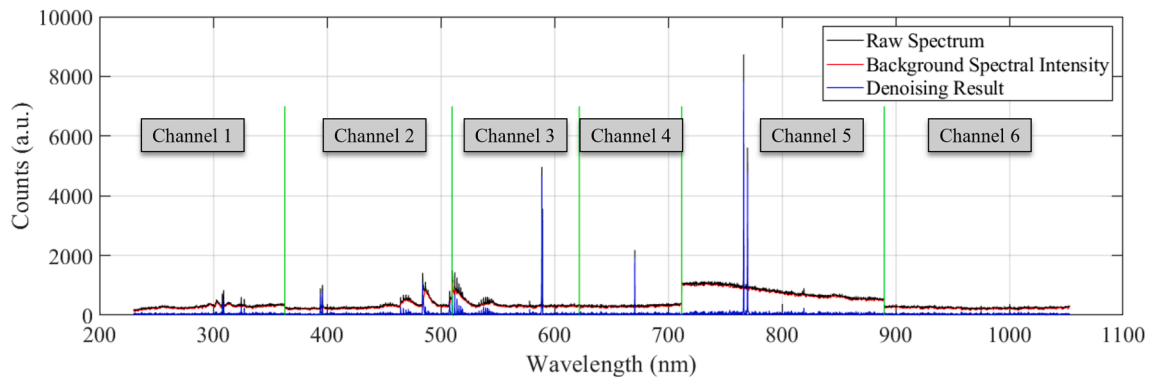


Fig. 6. Result of multi-channel spectral signal denoising based on secondary lower envelope separation.

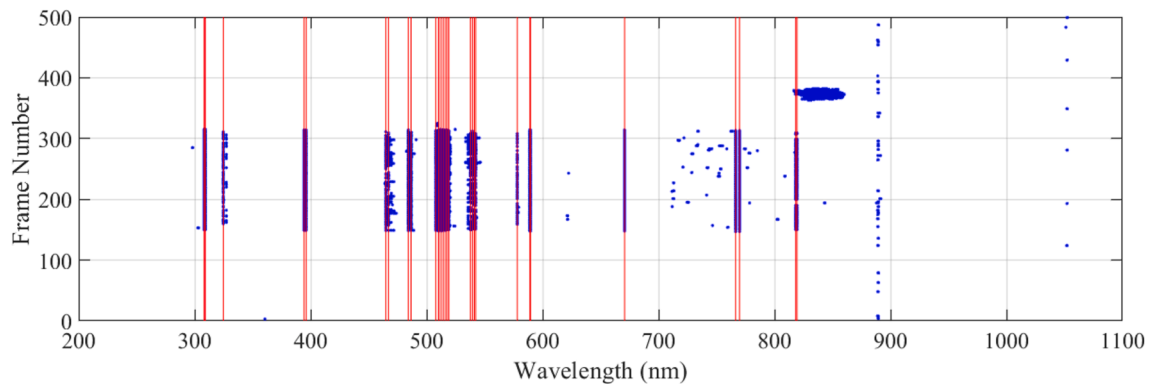


Fig. 7. The distribution of characteristic spectral lines.

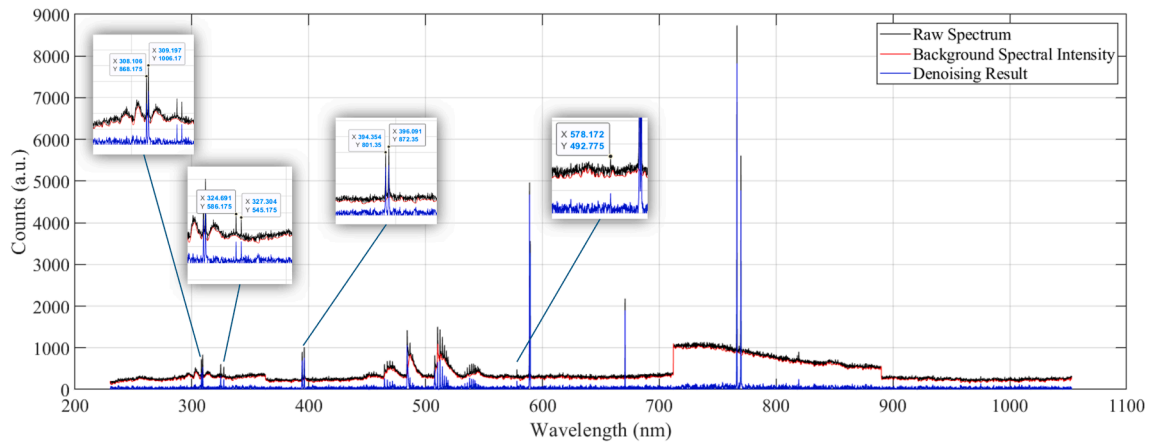


Fig. 8. Extracted characteristic spectral lines.

reference plane and the welding in-process keyhole depth detection reference plane relative to the welding head, causing a rotation error in the OCT measurement. As seen in the raw image from the OCT detection (Fig. 5), in the non-welding area, the keyhole depth values consistently remained above 0. To eliminate the welding head deviation error, a mean analysis was performed on the pre-welding and post-welding signal parts of each obtained OCT curve. Based on the analysis results, the OCT signal corresponding to the entire weld was corrected for welding head deviation error. The corrected OCT signal (Fig. 9) shows minimal fluctuations around 0 for the keyhole depth in the pre-welding and post-welding sections.

During the welding process, the morphology of the keyhole and its position relative to the laser change at an extremely high frequency.

OCT detection light may not always reach the lowest point of the keyhole, resulting in OCT signal fluctuations much larger than the actual fluctuation of the keyhole depth. The software provided with the Lessmuller device cannot effectively extract the fluctuation of the keyhole depth from the OCT signal (Fig. 10). Further processing of the OCT signal is needed to obtain a curve that closely reflects the actual fluctuation of the keyhole.

The depth information of the keyhole collected by OCT during the welding process exhibits typical non-linear characteristics. Empirical Mode Decomposition (EMD), as an adaptive signal time-frequency processing method based on Fourier transform, is particularly suitable for the analysis and processing of nonlinear and non-stationary signals. The EMD method repetitively computes mean envelope signals based on

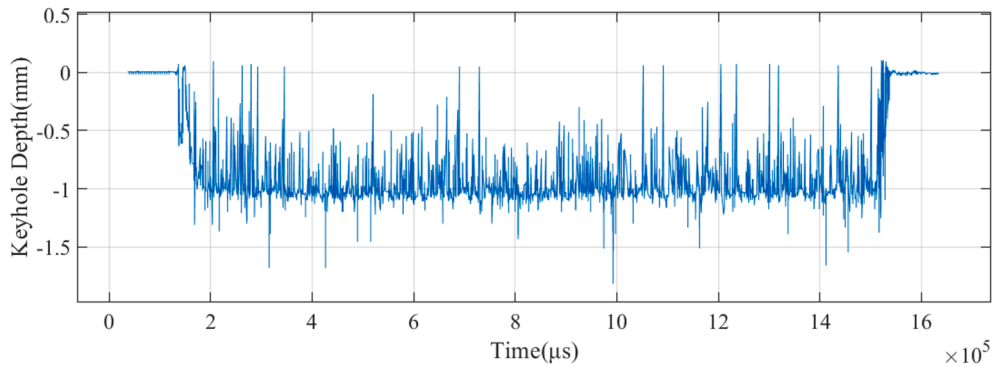


Fig. 9. The OCT signal laser deflection error elimination result.

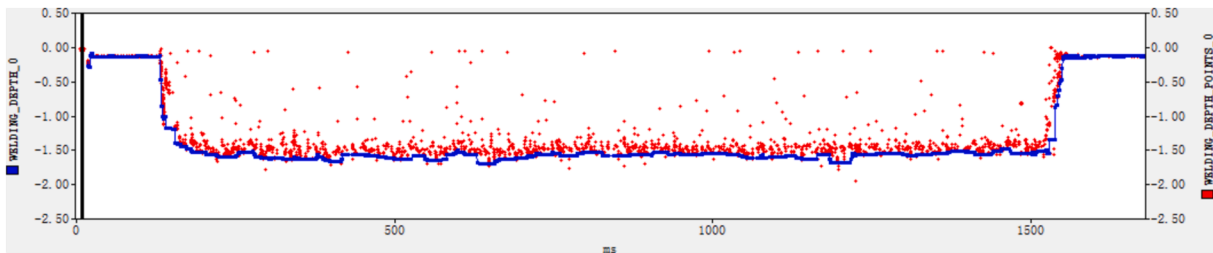


Fig. 10. The built-in software filtered out too many points, causing distortion in the pore depth fluctuation curve.

the upper and lower envelope lines of the signal until the signal satisfies two constraints: 1. The difference between the number of extremum points and the number of zero-crossings within the entire data segment is less than or equal to 1; 2. At any moment, the upper and lower envelope lines are symmetric to the time axis, i.e., the average value of the upper and lower envelope lines formed by local extremum points is 0.

Such a signal is called an Intrinsic Mode Function (IMF) of the original signal. By subtracting the obtained IMF from the original signal and repeating the above steps, other IMFs can be obtained. In this study, the original signal is decomposed into nine to ten IMFs, plus the final residue signal, resulting in ten to eleven sub-signals (Fig. 11 (a)). The sum of all signals is the original signal. By eliminating different IMFs, different keyhole depth fluctuation curves with varying levels of denoising can be obtained (Fig. 11 (b)). The first sub-signal is the sum of all IMFs and the residue, the second sub-signal is the sum of all IMFs and the residue after removing the first IMF, and so on. The initial IMFs in the EMD decomposition successfully separated high-frequency noise from the keyhole depth fluctuation signal, while the subsequent IMFs retained only the most macro-level fluctuations of the keyhole. Through the analysis of the Pearson correlation coefficient between the overall fluctuations of keyhole depth and the overall fluctuations of melt depth after removing different IMFs, it was found that the correlation coefficient is highest for the keyhole depth fluctuation curve after removing the first two IMFs (i.e., the third keyhole depth fluctuation sub-signal). Therefore, in this study, the third sub-signal of the keyhole depth fluctuation is chosen as a representative of the keyhole depth fluctuation.

4.3. SWI prediction of aluminum-copper dissimilar metal laser lap welding joint based on BPNN and multiple signals

Based on the aforementioned research, we have obtained spectral signals characterizing the melt volume ratio of upper and lower plates in laser lap welding of dissimilar aluminum-copper metals, as well as OCT signals characterizing the fluctuation of melt depth. However, in equation (3), there exists a hyperparameter (k), and its value shows significant non-linear variations under different process parameters. To achieve a more accurate prediction model for the SWI in laser lap

welding of dissimilar aluminum-copper metals, a BPNN-based prediction model was established using OCT and spectral signals [42]. In 45 experiments with 9 different process parameters, one experiment was randomly chosen as the test group for the BPNN model in each process parameter, while the others served as training data for BPNN. In this study, BPNN consisted of one input layer, three hidden layers, and one output layer. The input layer had 8 nodes (representing the intensity of 7 characteristic spectral lines and 1 OCT signal intensity), three hidden layers had 13, 13, and 7 nodes respectively, and the output layer had one node (representing the SWI of a single weld). The activation function was set to Sigmoid, and the target training accuracy was 0.0001 (Fig. 12). The training state, regression and performance of the BPNN training process is shown in Fig. 13. BPNN achieved the target accuracy in the training set at the 27,852nd iteration.

The performance of the BPNN was tested using a test set composed of experiments at nine different powers that were not included in the training set. The results indicated that the average error of the BPNN model for predicting the SWI of the lap joint in the test set was only 10.7 μm , with a maximum prediction error of only 19.4 μm occurring at a total power of 1920 W with adjustable annular beam (Fig. 14(a)). To verify the approach based on multi-signal monitoring of the lap joint SWI, attempts were made to train a BPNN model with OCT signals and spectral signals separately for the same structure. The SWI prediction model based on OCT had an average error of 47.8 μm in the test set (about 5 times that of the multi-signal prediction model) and a maximum error of 170.2 μm (about 10 times that of the multi-signal prediction model) (Fig. 14(b)). The SWI prediction model based on spectral signals had an average error of 41.6 μm in the test set (about 4 times that of the multi-signal prediction model) and a maximum error of 103.7 μm (about 5 times that of the multi-signal prediction model) (Fig. 14(c)). The results show that the predictive abilities of OCT signals or spectral signals for lap joint SWI were far inferior to those of the multi-signal prediction model, indicating that the model proposed in this study effectively combined the characteristics of the two signals and achieved high-precision prediction of the lap joint SWI in aluminum-copper dissimilar metal adjustable annular beam laser lap welding. The superior predictive performance of the multi-signal approach

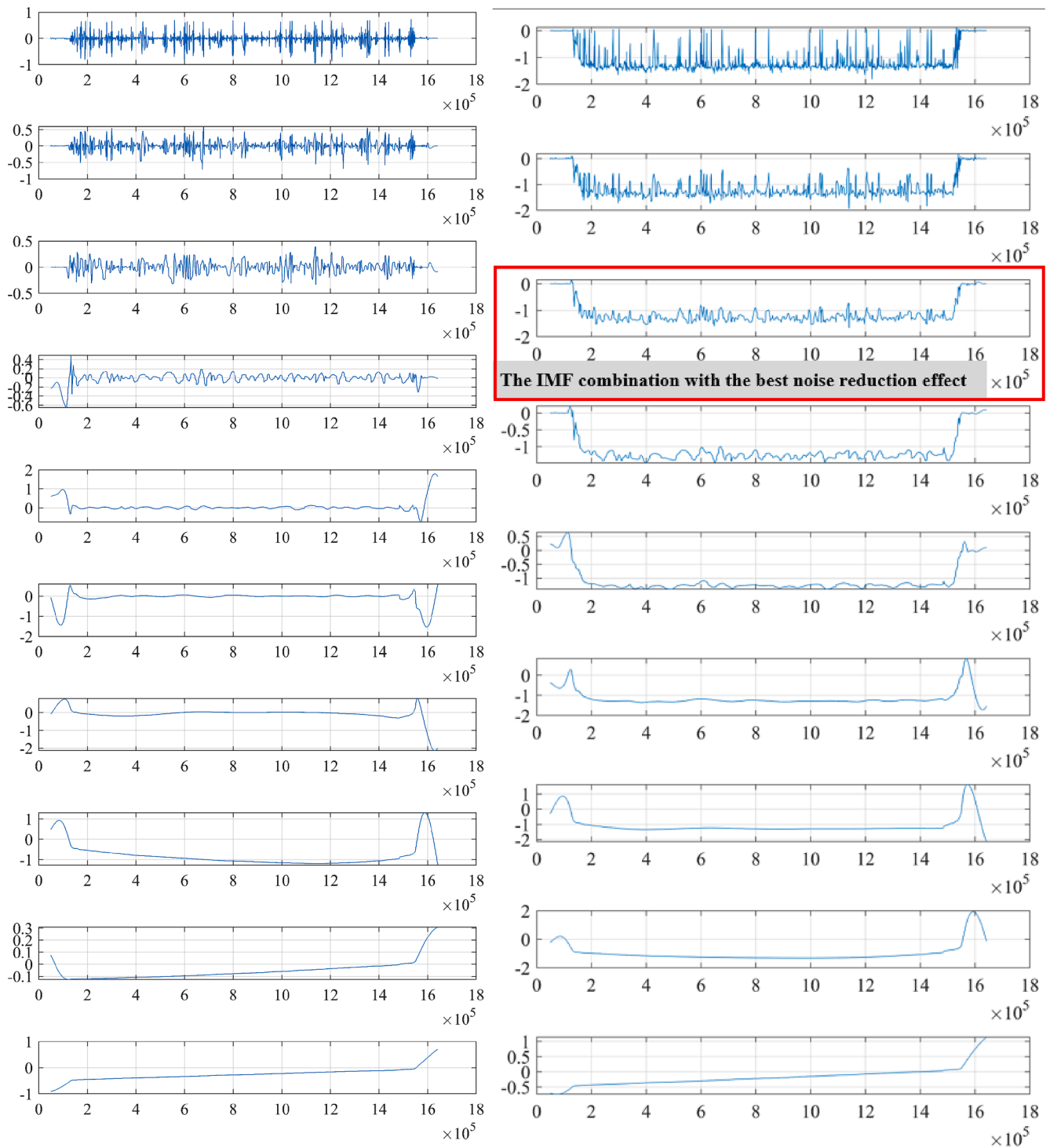


Fig. 11. (a) IMFs and error separated from the original OCT signal (b) Keyhole depth fluctuation curve after removing different IMFs.

mentioned in this paper compared to single-signal methods may be due to the fact that the SWI is influenced by both the two-dimensional depth fluctuations represented by the OCT signal and the three-dimensional molten volume information provided by the spectral signal. Single-signal prediction methods can only provide part of the information needed for detecting the SWI. The multi-signal monitoring approach allows for effective complementary use of these two types of information, achieving high-precision prediction of the SWI.

5. Conclusion

Prediction of SWI in aluminum-copper dissimilar metal lap joints provides essential insights for welding quality assessment. This paper proposes a novel method for high-precision prediction of SWI based on multi-signal and BPNN in aluminum-copper dissimilar metal ARM laser lap welding. The main conclusions are summarized as follows:

- 1) In laser lap welding of aluminum-copper dissimilar metals, the correlation coefficient between the melted volumes of the upper and

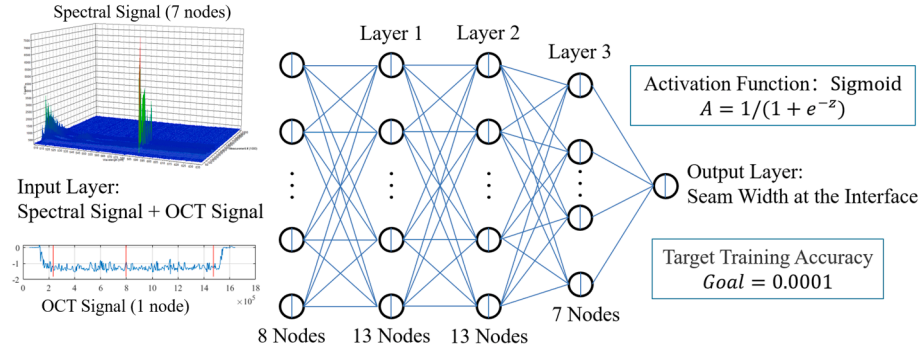


Fig. 12. BPNN structure and hyperparameters.

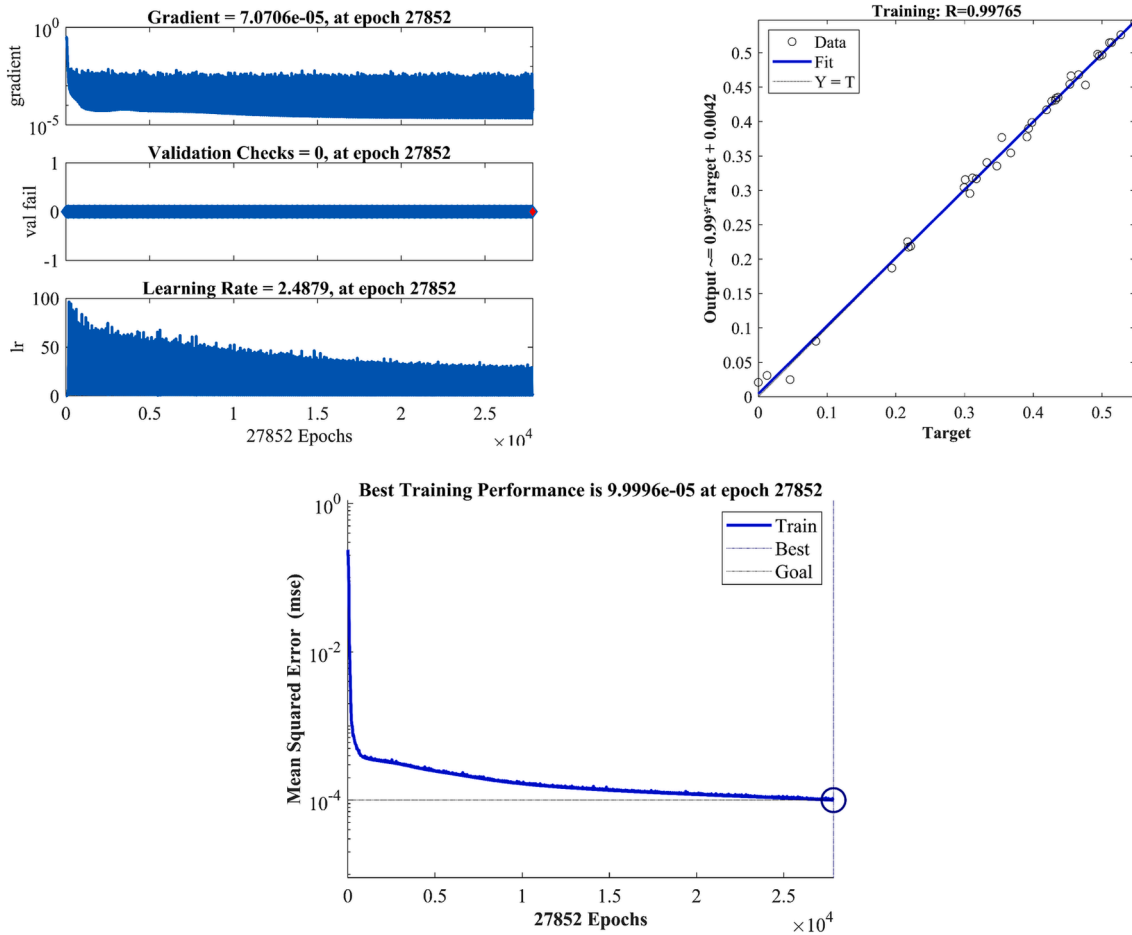


Fig. 13. The training state, regression and performance of the BPNN.

lower plates, the penetration depth and SWI reaches 0.9343. Based on these two features, a novel SWI characterization method was proposed, which confirms the feasibility of predicting SWI and provides a theoretical basis for subsequent quantized research.

2) The OCT signal provides keyhole depth information, serving as the basis for monitoring penetration depth fluctuations. Meanwhile, the intensity of certain characteristic spectral lines in the optical emission spectra of the laser induced plasma plume can support the monitoring of the upper-to-lower plate molten volume ratio. Combining these two signals, a method for online monitoring of the SWI in aluminum-copper dissimilar metal ARM laser lap welding is proposed, addressing the challenge of detecting this typical internal feature.

3) The SWI monitoring method based on multi-signal and BPNN proposed in this paper achieves an average error of only 10.7 μm , which is significantly lower than the prediction error of OCT or spectral signals alone, by almost an order of magnitude. This method achieves high-precision prediction of SWI in aluminum-copper dissimilar metal ARM laser lap welding.

The proposed method enables high-precision prediction of SWI in aluminum-copper dissimilar metal ARM laser lap welding. Future research will focus on collecting detailed data on the variation of SWI in single weld to establish a comprehensive dataset of SWI fluctuations and integrate monitoring of depth of fusion fluctuations to support detection of incomplete fusion defects in practical production process.

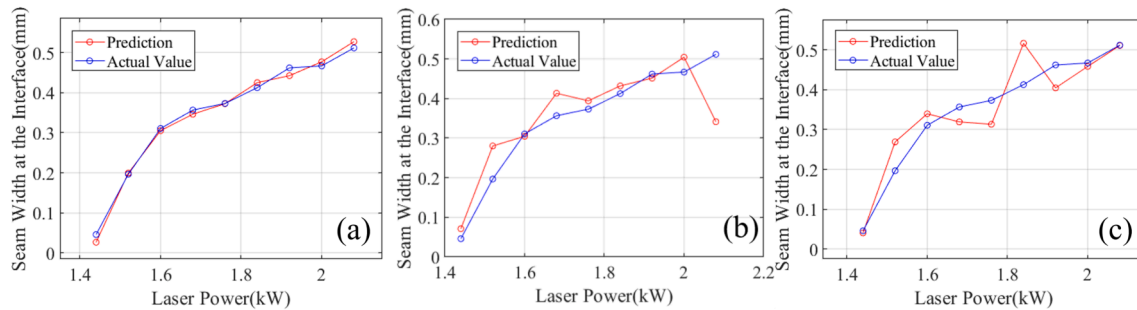


Fig. 14. (a) The prediction results of SWI based on OCT and spectral signals (b) The prediction results of SWI based on OCT signal (c) The prediction results of SWI based on spectral signal.

CRedit authorship contribution statement

Li Shixuan: Formal analysis, Investigation, Resources, Software, Writing – original draft, Writing – review & editing, Conceptualization. **Leshi Shu:** Writing – review & editing, Supervision, Methodology, Investigation. **Jiang Ping:** Writing – review & editing, Supervision, Methodology, Funding acquisition. **Jiang Shiliang:** Software. **Wu Wendi:** Conceptualization, Methodology. **Gao Yu:** Conceptualization, Software, Formal analysis. **Wang Yuan:** Conceptualization, Data curation, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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